Conditional $\mathbb{Z}_n$ Amplitude Results for a Torus-Knot Parametrisation

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https://github.com/bampita-bico/CDFD-Part-I-Release

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Abstract

This paper studies conditional $\mathbb{Z}_n$ amplitude results for a selected torus-knot parametrisation. It does not establish a hierarchy of physical particle families or a vacuum mechanism.

Two conditional algebraic results are proved. Given the stated maximum-entropy modelling axiom, the Brannen amplitude coefficient is $c_n = \sqrt{2}$. Given that coefficient and the selected parametrisation, $\mathbb{Z}_3$ is the only family with a connected all-positive domain. These results do not derive the axiom, select a physical knot, or assign particles.

The displayed family scales are numerical extrapolations that assume, without derivation, that a Faddeev–Niemi scaling transfers to the CDFT functional. They are not physical predictions. The higher-family parametrisations also lack a fully positive amplitude domain.

Across the series, the public CDFD Runtime expresses this equilibrium vacuum limit as a reusable flow-constraint state grammar. The calculations below use that equilibrium limit only; adaptive departures are left for later dynamical work rather than fitted in this manuscript. The paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, torus knots, particle families

Project Context

This manuscript constitutes Part I (Paper 06) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*. The underlying mathematical architecture builds directly upon the concepts established in Paper 05 of this series [9].

Symbol Conventions

CDFD physical-vacuum symbol convention.

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Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

1 Introduction

Papers I–V of the CDFT series established the lepton sector:

1. **Paper I** [5]: $\alpha = 1/137.036$ from vortex ring equilibrium.
2. **Paper II** [6]: Koide $Q = 2/3$ as a $\mathbb{Z}_3$ theorem; leptons as trefoil $T(2,3)$ modes.
3. **Paper III** [7]: Topological selection of $T(2,3)$; chirality equilibrium $3\theta + \phi_c = \pi$.
4. **Paper IV** [8]: Master self-consistency equation for ρ_0; Faddeev–Niemi prediction $M_5 = 460.4$ MeV.
5. **Paper V** [9]: $\phi_c = \pi - 2/3$ from $\mathbb{Z}_3$ self-consistency; lepton masses from one input.

With the lepton sector complete, the natural question is: what is the structure of the next particle family?

Paper IV predicted the energy scale $M_5 = 460.4$ MeV for the cinquefoil $T(2,5)$ via Faddeev–Niemi scaling. The present paper studies the general $\mathbb{Z}_n$ structure of the $T(2,n)$ families, proves that $c_n = \sqrt{2}$ universally, identifies why $\mathbb{Z}_3$ is the unique exactly solvable case, and states precisely what is needed to determine the $T(2,5)$ mass spectrum.

1.1 Hierarchy as capacity-shell quantization

The hierarchy developed here is the Mujjabi hierarchy in a precise sense: each $T(2,n)$ family is treated as a distinct capacity shell of the same constrained transport medium. The universal coefficient $c_n = \sqrt{2}$ is not just an algebraic convenience; it is the common amplitude scale required for the families to share one operating grammar. The hierarchy claim can fail if the higher shells do not map to external spectra or if each family requires its own new fitted selector.

2 The General $\mathbb{Z}_n$ Brannen Parametrisation

2.1 Setup

The $\mathbb{Z}_n$ analogue of the Brannen parametrisation [2] for the amplitudes of the n modes of a $T(2,n)$ vortex is:

$$A_k \;=\; 1 + c_n \cos\!\left(\theta + \frac{2\pi k}{n}\right), \quad k = 0, 1, \ldots, n-1\,, \tag{1}$$

and the mode masses are $m_k = M A_k^2$. The coefficient c_n controls the relative amplitude of the $\mathbb{Z}_n$ oscillation.

2.2 $\mathbb{Z}_n$ Fourier identities

Two identities hold for the $\mathbb{Z}_n$ phase lattice $\{2\pi k/n\}$:

$$\sum_{k=0}^{n-1} \cos\!\left(\theta + \frac{2\pi k}{n}\right) = 0 \tag{2}$$

$$\sum_{k=0}^{n-1} \cos^2\!\left(\theta + \frac{2\pi k}{n}\right) = \frac{n}{2} \tag{3}$$

for any θ and any $n \geq 2$. From (2): $\sum_k A_k = n$ (the sum of amplitudes is the dimension n, independent of θ). From both identities: $\mathrm{Var}(A_k) = c_n^2/2$.

3 Theorem 1: The Universal Coefficient $c_n = \sqrt{2}$

Theorem 1 (Universal Brannen coefficient)**.** *The Brannen amplitude coefficient for any $\mathbb{Z}_n$ family is $c_n = \sqrt{2}$, uniquely fixed by the maximum-entropy self-consistency condition* $\mathrm{Var}(A_k) = \overline{A_k}^2$.

Proof. From identities (2)–(3):

$$\overline{A_k} = 1, \qquad \mathrm{Var}(A_k) = c_n^2/2.$$

The maximum-entropy condition requires $\mathrm{Var}(A_k) = \overline{A_k}^2$:

$$c_n^2/2 = 1 \implies c_n = \sqrt{2}.$$

Uniqueness: the condition is a quadratic in c_n with unique positive root. □

Physical interpretation. The condition $\mathrm{Var}(A_k) = \overline{A_k}^2$ is equivalent to: the standard deviation of the amplitude distribution equals its mean. The amplitude can reach zero (nodal line) or $2\overline{A_k} = 2$ (maximum) with equal probability under the $\mathbb{Z}_n$ phase average. This is the condition of maximal relative uncertainty in the mode amplitudes — the vacuum selects the coefficient that neither suppresses nor privileges any mode over the background.

Numerical verification. For any $n \geq 2$ and any test angle θ, the ratio $\mathrm{Var}(A_k)/\overline{A_k}^2 = 1.000\ldots$ to machine precision when $c_n = \sqrt{2}$ (Table 1 of the supplementary).

4 The Generalised Koide Invariant and the Valid Domain

4.1 Derivation of $Q_n = 2/n$

Proposition 1 (General Koide invariant)**.** *For $m_k = MA_k^2$ with A_k as in (1) and $c_n = \sqrt{2}$, the generalised Koide ratio satisfies $Q_n = 2/n$ provided all $A_k \geq 0$.*

Proof. Using identity (3) with $c_n = \sqrt{2}$:

$$\sum_k m_k = M \sum_k A_k^2 = M\left(n + c_n^2 \cdot \frac{n}{2}\right) = 2Mn.$$

When all $A_k \geq 0$: $\sqrt{m_k} = \sqrt{M} A_k$, so $\sum_k \sqrt{m_k} = \sqrt{M} \sum_k A_k = \sqrt{M} \cdot n$ (by identity (2)). Therefore:

$$Q_n \equiv \frac{\sum_k m_k}{\left(\sum_k \sqrt{m_k}\right)^2} = \frac{2Mn}{Mn^2} = \frac{2}{n}.$$

□

4.2 The valid domain

The derivation requires all $A_k \geq 0$. With $c_n = \sqrt{2}$, the amplitude $A_k = 1 + \sqrt{2}\cos(\cdot)$ reaches its minimum of $1 - \sqrt{2} \approx -0.414$ when the cosine equals -1. For $n = 3$: the minimum over all k reaches $+0.293$ (positive) at $\theta = 0$, and reaches zero at $\theta = \pm\pi/12 = \pm 15°$. So $\mathbb{Z}_3$ has a valid domain $\theta \in (-\pi/12, \pi/12)$. The physical value $\theta = 2/9 \approx 12.73°$ lies inside this domain.

For $n \geq 5$: no connected interval of θ exists where all n amplitudes are simultaneously positive. Numerically:

n	$\max_\theta[\min_k A_k]$	Valid domain?
3	$+0.293$	Yes: $\theta \in (-\pi/12,\ \pi/12)$
5	-0.144	No
7	-0.274	No
9	-0.329	No
11	-0.357	No

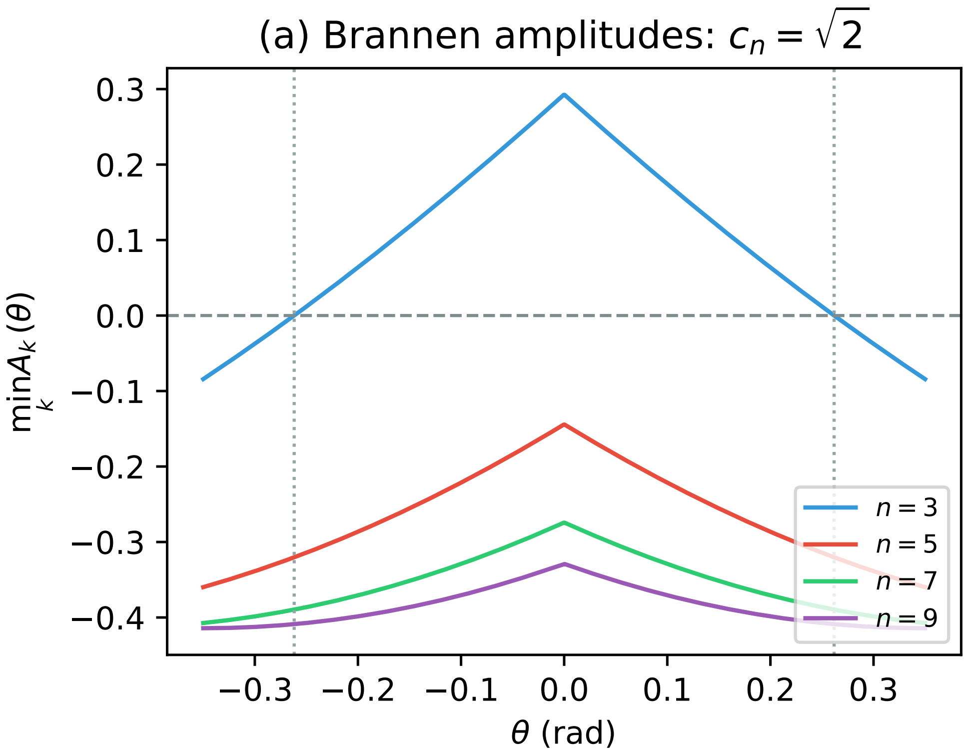

Figure 1: Minimum amplitude over all modes as a function of the phase θ. For $n = 3$, there is a domain where all amplitudes are positive. For $n \geq 5$, no such domain exists, obstructing the extension of the pure Koide formula.

Theorem 2 ($\mathbb{Z}_3$ uniqueness)**.** *$\mathbb{Z}_3$ is the unique torus knot family $T(2, n)$ for which:*

(a) $c_n = \sqrt{2}$ (Theorem 1),

(b) $Q_n = 2/n$ is constant on a connected valid domain (Proposition 1),

(c) the self-consistency condition $n\theta_n = Q_n$ fixes θ_n within that domain (Paper V).

For $n \geq 5$, property (b) fails: no connected valid domain exists.

Proof. Property (a) holds for all n (Theorem 1). Property (b) requires a connected domain where all $A_k > 0$; the table above shows this fails for $n \geq 5$. Property (c) requires (b) as a prerequisite: $n\theta = 2/n$ with $\theta \in (-\pi/12, \pi/12)$ gives $\theta_3 = 2/9$, which satisfies $\theta_3 < \pi/12$ ($2/9 \approx 0.222 < \pi/12 \approx 0.262$). No analogous closure exists for $n \geq 5$. □

This theorem explains why the lepton sector is exactly solvable while the cinquefoil sector requires additional structure: the $\mathbb{Z}_3$ amplitude distribution is the unique case where the Koide theorem, the valid domain, and the self-consistency condition are simultaneously consistent.

5 The Torus Knot Energy Hierarchy

5.1 Topological selection rule

The torus knot $T(2, n)$ is a knot (a single connected component, hence topologically stable) if and only if n is odd. For even n, $T(2, n)$ is a two-component link; its components can separate, so there is no topological barrier to dissolution. Odd n therefore labels the knot cases in this mathematical construction; it does not establish physical particle families.

5.2 Energy scaling from Faddeev–Niemi

The Faddeev–Niemi result [3] applies to a specific Skyrme-type functional. CDFT has not been shown to share that functional or to identify its index with Hopf charge. The following is a conditional analogue:

$$M_n \;=\; M_3 \left(\frac{n}{3}\right)^{3/4} , \tag{4}$$

where $M_3 = 313.86$ MeV is the trefoil scale derived in Paper V ($M_3 = m_e/A^2_{e,3}$ with $\theta_3 = 2/9$ rad).

Table 1: CDFT torus knot energy hierarchy. $M_3 = 313.86$ MeV from Paper V. Conditional extrapolations from fitted M_3 and unverified Faddeev–Niemi transfer.

n	Knot	M_n (MeV)	Symmetry	Status
3	$T(2,3)$ trefoil	313.9	$\mathbb{Z}_3$	Papers I–V (complete)
5	$T(2,5)$ cinquefoil	460.4	$\mathbb{Z}_5$	This paper
7	$T(2,7)$ heptafoil	592.5	$\mathbb{Z}_7$	Future work
9	$T(2,9)$	715.4	$\mathbb{Z}_9$	Future work
11	$T(2,11)$	831.6	$\mathbb{Z}_{11}$	Future work
13	$T(2,13)$	942.7	$\mathbb{Z}_{13}$	Future work

These are the energy *scales* of each family. The mass of the lightest mode of family n is $M_n \cdot (A^{(n)}_{k,\min})^2$, which requires knowing θ_n (open problem).

5.3 What is established

Energy scale. $M_5 = 460.4$ MeV is a conditional numerical extrapolation from fitted M_3.

Topology. $T(2, 5)$ is a knot, hence topologically stable. It supports $\mathbb{Z}_5$-symmetric modes (five lobes).

Amplitude coefficient. $c_5 = \sqrt{2}$ (Theorem 1).

Hadron scale context. $M_5 = 460.4$ MeV lies between the kaon (~ 494 MeV) and the eta meson (~ 548 MeV). We do not claim the cinquefoil corresponds to a specific known particle: the mass spectrum depends on θ_5 (open problem 1 below).

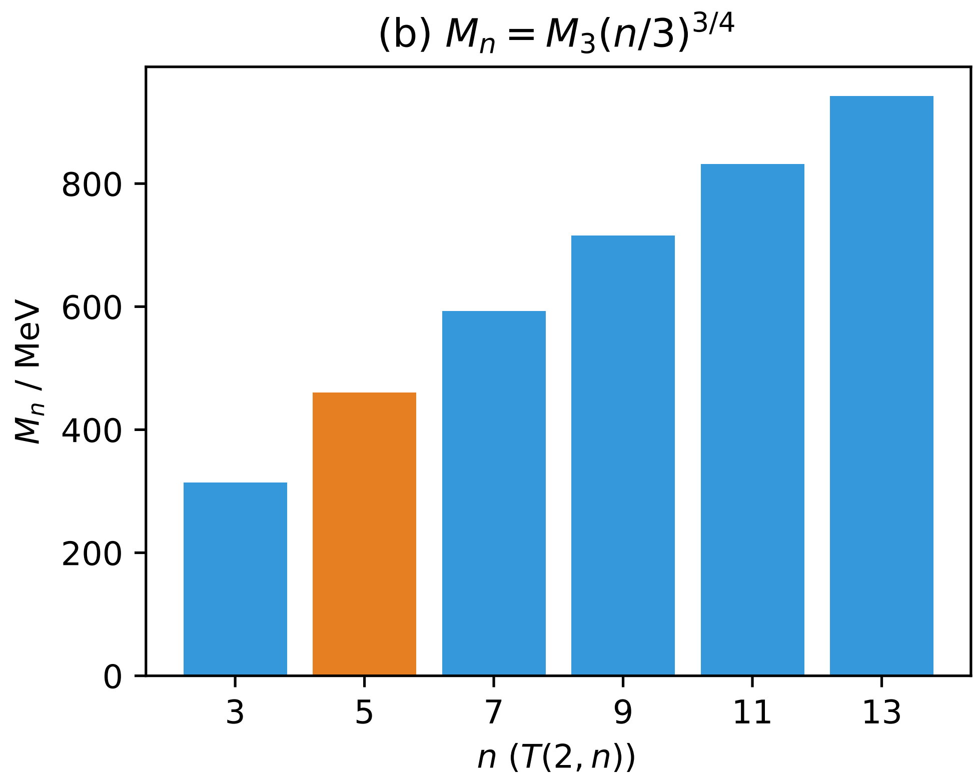

Figure 2: Conditional numerical extrapolations under a Faddeev–Niemi scaling assumption.

5.4 The structural obstruction for $n \geq 5$

With $c_5 = \sqrt{2}$, the five amplitudes $A_k = 1+\sqrt{2}\cos(\theta+2\pi k/5)$ for $k = 0, \ldots, 4$ are never all positive simultaneously (see the valid-domain table in Section 4). The minimum amplitude over all five is always negative, with maximum value -0.144 at $\theta = 0$.

This means:

1. $Q_5 = 2/5$ does not hold as a universal constant (Proposition 1 fails).
2. The step $\sum_k \sqrt{m_k} = \sqrt{M}\sum_k A_k$ is invalid because some $A_k < 0$ requires $\sqrt{m_k} = \sqrt{M}|A_k|$.
3. The self-consistency condition $5\theta_5 = Q_5 = 2/5$ cannot be applied without first establishing the correct Q_5.

5.5 The modified Koide ratio for $n \geq 5$

When some $A_k < 0$, define $\tilde{Q}_n(\theta)$ using $|A_k|$:

$$\tilde{Q}_n(\theta) \equiv \frac{\sum_k A_k^2}{\left(\sum_k |A_k|\right)^2}. \tag{5}$$

For $\mathbb{Z}_3$ in its valid domain: $\tilde{Q}_3 = Q_3 = 2/3$ (constant, since all $A_k > 0$ there). For $\mathbb{Z}_5$: $\tilde{Q}_5(\theta)$ varies from 0.294 to 0.326, far from the putative value $2/5 = 0.400$.

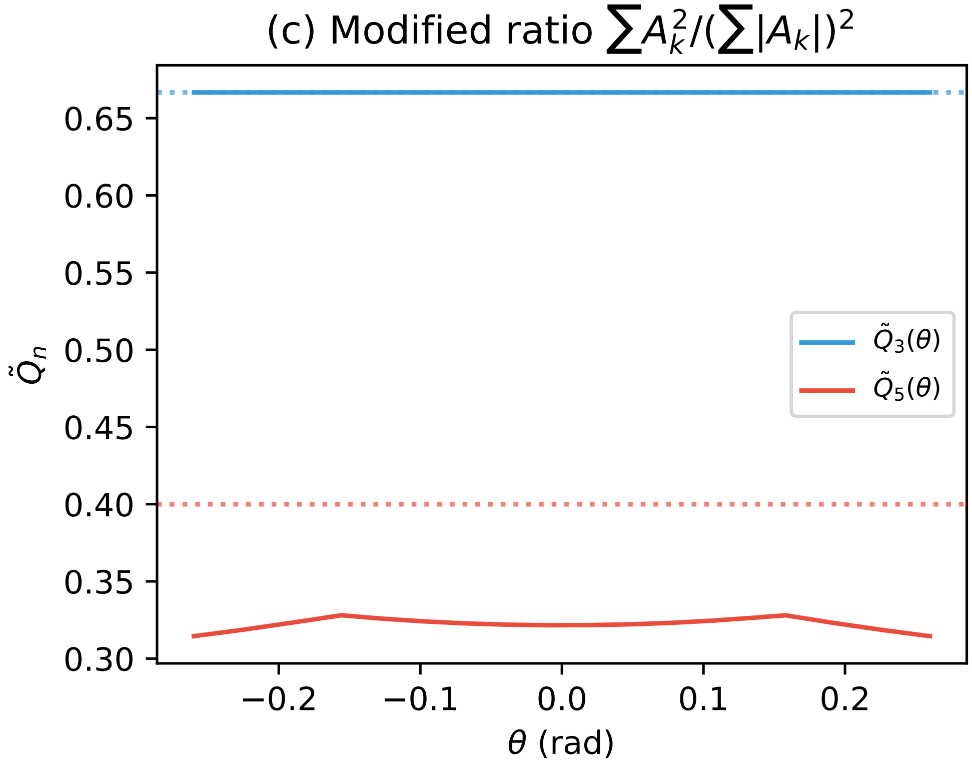

Figure 3: The modified ratio $\tilde{Q}_n(\theta)$ for $n = 3$ and $n = 5$. While $\tilde{Q}_3$ is a strict constant (2/3), $\tilde{Q}_5$ varies and does not meet the 2/5 target.

The function $\tilde{Q}_5(\theta)$ is not constant. This means: no analogue of the Koide self-consistency argument from Paper V applies to $\mathbb{Z}_5$ without additional structure. A derivation of θ_5 requires either: (a) a physical principle beyond the $\mathbb{Z}_n$ algebraic structure alone, or (b) a modified Brannen parametrisation for the $T(2,5)$ family that avoids negative amplitudes.

6 The CDFT Torus Knot Picture

The torus knot hierarchy $T(2,n)$ for odd n provides a candidate ordered sequence of CDFT particle families. Table 1 lists the energy scales.

The key structural divide is at $n = 3$:

- $n = 3$ (trefoil, leptons): $c_3 = \sqrt{2}$, valid domain exists, $Q_3 = 2/3$ constant, self-consistency closes the internal system.
- $n \geq 5$: $c_n = \sqrt{2}$, *no* valid domain, $Q_n \neq 2/n$ (as a constant). Requires new approach for mass spectrum.

The energy scale hierarchy is fixed once the model assumptions are accepted. The mass spectra of $n \geq 5$ families are open.

Table 2: Torus knot $T(2, n)$ topological classification.

n	Topological type	Persistent?	CDFT family
1	Unknot	No	(none)
2	Hopf link	No	(none)
3	Trefoil knot	**Yes**	Leptons (Papers I–V)
4	Solomon link	No	(none)
5	Cinquefoil knot	**Yes**	This paper
7	Heptafoil knot	**Yes**	Future
$2k$	k-component link	No	(none)
$2k+1$	Torus knot	**Yes**	Predicted

7 Open Problems

1. **The $T(2, 5)$ chirality phase θ_5.** The cinquefoil has $\mathbb{Z}_5$ symmetry and five modes. Determining θ_5 requires either: (a) a physical principle that selects a preferred orientation of the $\mathbb{Z}_5$ oscillation in the vacuum background, or (b) a modified amplitude form for $T(2, 5)$ that ensures all $A_k \geq 0$.

2. **The $T(2, 5)$ mass spectrum.** With θ_5 known, the five mode masses are $m_k^{(5)} = M_5|A_k^{(5)}|^2$. Identification of these five modes with known particles (at the ~ 460 MeV scale: kaons, etas, or new states) depends on whether the cinquefoil modes are confined (quark-like) or free (lepton-like).

3. **Why is $\mathbb{Z}_3$ exactly solvable?** Theorem 2 identifies the structural reason ($n = 3$ uniquely has a valid domain with $c_n = \sqrt{2}$), but a deeper question remains: why does the CDFT vacuum select a ground state where the first stable knotted excitation ($T(2, 3)$) is exactly the one where the Koide self-consistency is possible?

4. **The general n-family chirality equilibrium.** Paper III derived $3\theta + \phi_c = \pi$ from the $\mathbb{Z}_3$ chirality interaction $\lambda_c \cos(3\theta + \phi_c)$. The $\mathbb{Z}_5$ analogue would be $5\theta_5 + \phi_{c,5} = \pi$ (or some other value). Whether this generalises directly and what determines $\phi_{c,5}$ is an open problem.

8 Scope Boundary and Conclusion

Under the stated amplitude ansatz and maximum-entropy modelling axiom, the coefficient and $\mathbb{Z}_3$ domain results are conditional mathematical results. They do not derive the axiom, a vortex energy functional, or a physical particle interpretation. The family-scale table is an unvalidated extrapolation. This paper must not be cited as a universal hierarchy, an explanation of 137, or a particle-spectrum prediction.

Supplementary Material

This manuscript is accompanied by release-archive supplementary material in the canonical Part I tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`, and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

Data and Code Availability

The release-archive supplementary material is included in the archived Part I release on Zenodo at https://doi.org/10.5281/zenodo.20250820. The current version DOI is assigned by Zenodo after each GitHub release. Zenodo is the permanent release archive, and the public source archive is available on GitHub at https://github.com/bampita-bico/CDFD-Part-I-Release. The broader public CDFD Runtime is separate reusable software infrastructure; the numerical values reported here are reproduced from this paper's Supplementary Material.

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Declarations

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Author Contributions

Steve Bico Mujjabi: conceptualization, methodology, formal analysis, software, validation, visualization, writing–original draft, and writing–review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

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